

INSTITUTE OF ACTUARIES OF INDIA

Subject CB1 – Business Finance

November 2025 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution. 1) Prepaid expenses	E	(2)
Solution. 2) Commercial Papers are repaid with face value plus agreed interest rate which usually exceeds Government securities	B	(2)
Solution. 3) Decrease Tax Credits on the asset	A	(2)
Solution. 4) Horizontal Analysis	B	(2)
Solution. 5) iv), i), iii), ii)	B	(2)
Solution. 6) Sale of Inventory on Account	B	(2)
Solution. 7) It is not a charge against profits but an appropriation of profits	D	(2)
Solution. 8) Cost of Debt	B	(2)
Solution. 9) 18 th July	C	(2)
Solution. 10) Survival and self-sufficiency	C	(2)

Solution 11:

- a) The beta of a stock is a measure of the **systematic risk** of the stock relative to a **diversified portfolio of all risky assets** (i.e. the market).

The market would have a beta of 1. (2)

- b) The beta of the company may be measured by:
- Looking at the **company's historical returns** on equity and **comparing against market returns**, but subject to significant variation for example dependent upon the period considered
 - Considering the **industry beta** based on a **range of companies undertaking similar activities**
 - Estimating**, based on **knowledge of the company and its industry** and how the industry might **react to changes in the market** (3)

- c) A stock with a beta of 1 implies that the stock behaves in line with the market. Depending on expectations of market movements, as an aggressive investor, **may wish to invest in stocks with higher positive beta to maximize short-term gains.** A stock with a negative beta of 1 signifies that the stock behaves opposite to the market.

If the market index improves, the stock will lose value and vice-versa. **Is a downturn in the market is expected, low beta stocks may be attractive in the short to medium term.** (3)

- d) Cash (1)
- [9 Marks]**

Solution 12:

- a) Payback Period—
 Project A: 1 Year.
 Project B: $1 + 2500/7500 = 1.33$ Years.
 Project C: $2 + 4000/12000 = 2.33$ Years.
 Project D: 1 Year.

Ranking: A, D, B, C. (2)

- b) IRR—
 Project A: There are no returns as IF = OF and hence IRR = 0

Project B: Even cash flows, hence $PVAF = 10,000 / 7,500 = 1.33$ for 2 years, IRR = Approx 32.22% for 2 Yrs

Project C: Trial Error Method NPV@25% = +304 & @30% = -633 and IRR = 26.62%

Project D: Trial Error Method NPV@35% = +273 & @40% = -233 and IRR = 37.70%

Ranking: D, B, C, A.

(6)

c)

Ranking as per 10%: C, D, B, A.

Ranking as per 30%: D, B, C, A.

(4 marks with calculation)

[12 Marks]

Solution 13:

The firm's Plan for growth can be constrained for several reasons:

1) Difficulty in raising Finance:

Investors will consider the expected outlook for the economy in general, and the **firms' investment plans, its reputation and its creditworthiness.**

2) Fears of a fall in the share price:

If firm doesn't have confidence of its shareholders, a fall in dividends because of profit being invested in the business might cause shareholders to sell their shares.

The fall in share can lead to **takeover bid** as well.

3) Lack of Managerial experience and expertise:

Running large business is more complex, more time **consuming and more expensive** than running a small business. It requires skilled managers, clear management structures and delegation.

Some owners/managers of small business may be reluctant to lose direct control and might resist growth or manage it badly.

4) Limited time to prepare the workforce including management.

<u>Cashflow (In Rs.)</u>	A	B	C	D
<u>NPV@10%</u>	-909	3,016	4,139	3,824
<u>NPV@30%</u>	-2,307	207	-633	833

5) Government policy on monopoly power and mergers.

Government may wish to prevent takeovers on the grounds of national security or monopoly, when the domestic company is purchased by an overseas competitor.

The competitor mayn't have the same loyalty to the country so may move jobs/head office functions overseas for efficiency.

However, it may be impossible to prevent takeovers, or even to try to prevent them as to do so may breach international rules.

[5 Marks]

Solution 14:

a) Current Ratio will be defined as **Current Assets divided by Current Liabilities.**

This Ratio is used to assess whether the company will be able to pay its bills over the next few months.

Quick Ratio also known as acid / liquidity ratio aims at looking at **short -term liquidity.**

Quick Ratio can be defined as **(Current Assets-Inventories- Prepaid expenses)/Current Liabilities**

(2)

b) Current Ratio = 3.5: 1 Quick Ratio = 2: 1

Let Current liabilities = x

Current assets = 3.5x and

Quick assets = 2x

Inventories = Current assets – Quick assets

24,000 = 3.5x – 2x

24,000 = 1.5x

Current Liabilities = Rs. 16,000

Current Assets = 3.5x = 3.5 × Rs. 16,000 = Rs. 56,000. (5)

Verification:

Current Ratio = Current assets: Current liabilities

= Rs. 56,000: Rs. 16,000

= 3.5: 1

Quick Ratio = Quick assets: Current liabilities

= Rs. 32,000: Rs. 16,000 = 2: 1

[7 Marks]

Solution 15:

The key considerations from the CFO's commentary is as follows

1. There are two advantages to the bondholders, one is **regular interest payments** and a potential **capital gain** from the warrant.
2. The **capital gain** will arise from the issue of fresh shares at the exercise price stated in the warrant, so the company does **not need to find any cash** to make the payment.
3. This **contrasts with the cash flows** that are associated with paying **interest and capital on a traditional bond**.
4. If the warrant is **exercised**, then the existing shares in issue will be **diluted**.
5. In other words, the **market value** of each share will **fall** in response to the discount implicit in the exercise price relative to the market value at exercise.
6. The warrant may **never be exercised**, in what case there will be **long term cost** to the entity or the shareholders. (5 marks)

Solution 16:

1. Equity Financing

- **Equity Shares**: are the most important **source of long-term capital** for the company.
- They represent **ownership in** the company and therefore, the funds raised through their issue are called owner's funds.
- Shareholders do **not receive fixed dividend**. **Dividends** are **paid** to the shareholders based on the **company's earnings**.
- In addition, the shareholders have **voting rights**, which allow them to **participate** in the **management and decision-making** of the company
- **Preference shares**- Preference shareholders receive **fixed dividends regardless** of company's profitability, unlike equity shareholders who receive **dividends based on company's performance**.
In the event of company **liquidation**, preference shareholders are **paid back their capital before equity shareholders**.

While some preference shares may have **limited voting rights**, most **do not have** the right to participate in company **decision-making**.

Some preference shares **are redeemable**, meaning the company can buy them back at a predetermined price after a specified period.

2. Debt Financing

- **Debentures:** It is an **acknowledgment by the company** that the company has borrowed certain amount from the debenture holders.
- The **interest rate is predetermined and fixed at the time of issue**. The interest rate must be **paid whether the company makes a profit or not**.

(maximum 1 marks)

- **Bank Loans and Credit**

Obtaining loans from **commercial banks or financial institutions**, at a **fixed or variable rate** of interest.

Other Sources

- **Retained earnings:** Profits are **saved and reinvested** in business for expansion.
- This may lead to **shareholder dissatisfaction** due to **lower dividend payouts**.

The company may have **large**, retained earnings but **low cash as profits may be tied up in assets** such as inventory.

(7 Marks)

Solution 17:

1. Offering **discounts** towards the year end to **encourage customers to enter contracts sooner** than they might do otherwise. Profits for the current year would **boost at the expense of the next year**.

2. Companies exploit loopholes in accounting standards to engage in **creative practices** that report higher profits.

3. Management can **bias estimates and assumptions** to make the published figures appear more impressive.

For example- **closing inventory** has to be valued at the **lower of cost and net realisable value**, with profit being affected by this valuation. An optimistic assumption about the net realisable value of every item in the inventory will boost profit.

4. Profitability figures can be **enhanced by timing major capital transactions**. Delaying the **acquisition of non-current assets** may reduce capital employed by enough to more compensate for the reduction in activity due to weaker asset base.

5. **Misleading disclosures**- omitting or presenting incomplete notes, such as significant **accounting policies, related -party transactions or contingent liabilities** can mislead readers about the true profitability.

[6 Marks]

Solution 18:

- a) Opportunity cost is the cost of not taking up the next best investment choice when making a financial decision.

It is the return forgone on alternate investments of equivalent risk.

(1)

b)

- If a company has several **mutually exclusive projects**, then the opportunity cost of **accepting one project** is that it will be **impossible to accept** any of the others.
- Companies might also **face capital rationing**. Accepting a positive NPV project might involve using **funds** that could be **applied more profitably elsewhere**.
- Considering opportunity costs make decision making little complex because of the need to determine the NPV of competing projects.
However, it may be difficult to spot opportunities that have been missed or overlooked and so there may be very little risk of criticism arising from a missed opportunity.

(4)

c) A company can't use the same opportunity cost for all the projects.

(2)

[7 Marks]**Solution 19:****Statement of Profit and Loss Account of Harish Chemicals Ltd.**

Particulars	In Cr.	
Revenue from Operations	42,68,000	
Cost of stock used:		
Cost Of material consumed	-23,19,000	
Change in inventory of finished goods	1,43,000	
Wages	-9,00,000	
Gross Profit		11,92,000
Other Expenses (Note: 9)	-4,76,000	
Operating Profit		7,16,000
Profit before interest and Taxation		7,16,000
Finance Cost	-1,71,000	
Other Income	56,000	
Profit Before Taxation		6,01,000
Tax		-
Profit for the year		6,01,000

Balance Sheet of Harish Chemicals Ltd. as at 31 March 20X1

Assets	Rs.
Non-Current Assets	
PPE	30,05,000
Good will	2,65,000
Current Assets	
Inventories	8,23,000
Trade receivables	4,40,000
Cash and Bank Balances	53,000
Short term Loans and advances	80,000
Total Assets	46,66,000

Equity and Liabilities	Rs.
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Shareholders' funds:	
Share Capital	25,00,000
Reserves and Surplus	7,40,000
Non-Current Liabilities	
Long Term Borrowing	11,45,000
Current Liabilities	
Trade Payables	2,81,000
Total equity and liabilities	46,66,000

Notes to Accounts:

1) Share capital

Capital (Shares of Rs.10 each)

25,00,000

2) Reserves and Surplus

Balance as per last Balance sheet

1,39,000

Balance in Profit and Loss Account

6,01,000

7,40,000

3) Long Term Borrowings

11% Debentures

5,00,000

Bank Loans (assumed long-term)

6,45,000

11,45,000

4) PPE

Freehold Land

15,46,000

a) Tools and Equipment

2,45,000

Furniture

2,00,000

Fixtures

3,00,000

22,91,000

b) Plant and Machinery

8,60,000

less: depreciation

-1,46,000

7,14,000**Total (a +b)****30,05,000**

5)

Cash and Bank Balances

Current Account

45,000

Cash

8,000

Total**53,000**

6) Other Income

Rent received

46,000

Transfer fees

10,000

Total**56,000**

7) Change in inventory

Opening

6,80,000

Closing

-8,23,000

Total**-1,43,000**

8) Finance Cost

Interest on Bank Loans

1,16,000

Debenture interest

55,000

Total	1,71,000
9) Other Expenses	
Consumables	84,000
Preliminary expenses	10,000
Bad debts	35,000
Discount	40,000
Rentals	25,000
Commission	1,20,000
Advertisement	20,000
Dealers' aids	21,000
Transit Insurance	30,000
Trade Expenses	37,000
Distribution freight	54,000
Total	4,76,000

[22 Marks]
